

Differentially Private Prompts Warehouse

1 Introduction

Utilizing differential privacy mechanisms to perturb tokens in the prompts has become a promising method for privacy protection in large language models. A series of works [YDW⁺21, CMW⁺23, TCZ⁺25, WDLY25] perturb each (sensitive) token by perturbing it to another with a specially designed exponential mechanism. This requires huge computational overhead to protect one prompt given a privacy budget. Consider a prompts warehouse keeps many private prompts. There are users who ask for some prompts with special privacy budgets.

Strawman Solution 1. Compute the private prompts with given budget on-the-fly, which leads huge computational overhead and long response time.

Strawman Solution 2. Pre-compute several private prompts with default budget (i.e., $\varepsilon = 1, 2, 4, 8, \dots$) and return one of perturbed prompts whose budget smaller than given one. Since smaller budget leads to higher protection but lower accuracy, this method leads to poor utility.

We aim to design a mechanism achieving both performance and utility.

2 Background

2.1 Exponential Mechanism

For any $\varepsilon > 0$, a randomized exponential mechanism $\mathcal{R}_{\text{EM}} : \mathcal{X} \rightarrow \mathcal{Y}$ satisfies ε -DP if, for each input $x \in \mathcal{X}$, the mechanism outputs $y \in \mathcal{Y}$ with probability:

$$\Pr[\mathcal{R}_{\text{EM}}(x) = y] \propto \exp\left(\frac{\varepsilon}{2\Delta} \cdot u(x, y)\right),$$

where $u(x, y)$ is a user-defined utility function and $\Delta := \max_{x \sim x', y} |u(x, y) - u(x', y)|$ is its global sensitivity.

2.2 Prompt Perturbation via DP

The naive DP mechanism works as follow

1. Divide the prompt into tokens $(x_1, x_2, \dots, x_n)$, where $x_i \in \mathcal{V}$ the vocabulary;
2. For each (sensitive, which is defined by a subset $\mathcal{V}_S \subseteq \mathcal{V}$) token x_i , compute its token embedding distance between all tokens $t \in \mathcal{V}$, denoted as $d(x_i, y)$;
3. Compute the utility function $u(x_i, y) \leftarrow f(d(x_i, y))$;
4. Use EM mechanism to sample a perturbed token and output it.

2.2.1 SanText

SanText¹ adopts the geometry of differential privacy (i.e., metric local differential privacy): Given $\varepsilon > 0$ and a distance metric $d : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$, for any $x \sim x'$, the mechanism satisfies ε -GP (or $\varepsilon \cdot d(x, x')$ -DP) if

$$\Pr[\mathcal{M}(x) = y] \leq e^{\varepsilon \cdot d(x, x')} \cdot \Pr[\mathcal{M}(x') = y].$$

This can be achieved by outputting token y with probability

$$\Pr[\mathcal{R}(x_i) = y] = C(\varepsilon, x_i) \cdot \exp\left(-\frac{\varepsilon \cdot d(x_i, y)}{2}\right),$$

where $C(\varepsilon, x_i) = \sum_{y \in \mathcal{V}} \exp(-\frac{\varepsilon d(x_i, y)}{2})$. To see it satisfies ε -GP, note that for any $y \in \mathcal{V}$,

$$\frac{\exp(-\frac{\varepsilon d(x_i, y)}{2})}{\exp(-\frac{\varepsilon d(x'_i, y)}{2})} \leq \exp\left(\frac{\varepsilon |d(x'_i, y) - d(x_i, y)|}{2}\right) \leq \exp\left(\frac{\varepsilon d(x_i, x'_i)}{2}\right)$$

¹<https://github.com/xiangyue9607/SanText>

Also note that any ε -GP mechanism is immediately $\varepsilon \cdot d_{\max}$ -DP, where $d_{\max} := \max_{x, x' \in \mathcal{V}} d(x, x')$.

SanText⁺ partitions the input to sensitive/non-sensitive sets $\mathcal{V} = \mathcal{V}_S \cup \mathcal{V}_N$ and the output to (un)protected sets $\mathcal{V} = \mathcal{V}_P \cup \mathcal{V}_U$. A sensitive input $\mathcal{V}_S$ is guaranteed a protected output $\mathcal{V}_P$. A non-sensitive input may lead to a protected or unprotected output, depending on the random coin.

2.2.2 CusText

In CusText², “Specifically, we pick an arbitrary unmapped token $x \in \mathcal{X}$, find the K tokens semantically closest to x , generate an K -to- K mapping from all the K tokens to themselves.” That is, CusText defines several disjoint groups of vocabulary of size K , and for each group T , it performs the general EM over $\mathcal{X} = \mathcal{Y} = T$.

2.2.3 InferDPT

In InferDPT³, the mapping is random: a random vector $\hat{x}$ is generated for the input x . Any output y whose $d(x, y) < d(x, \hat{x})$ is considered “close”, and EM is performed on “close” vectors for each input x .

2.2.4 Cape

In Cape, the authors consider “contextual logits” in addition to token distances, therefore defining the utility function as

$$u(x_i, y) = L(y \mid x_1, x_2, \dots, x_{i-1}) \cdot \exp(-D(x_i, y))$$

where L is the loss given by a client-side model. Depending on the model used, L may also depend on subsequent tokens $(x_{i+1}, \dots, x_n)$. D is the normalized euclidean distance.

Given the utility function, Cape runs EM with a bucketing trick: for x_i , it sorts $u(x_i, y)$ for all y ’s, and organizes them into N_b buckets where each bucket has a utility range of $(u_{\max} - u_{\min})/N_b$. y can then be sampled by drawing a bucket through EM, followed by drawing a random element from the bucket.⁴

3 Idea

3.1 Findings

All prior works on prompt perturbing under DP are based on the exponential mechanism. In addition, there are two key properties that are specific to this application:

1. Any pair of (sensitive) tokens are considered neighbors, i.e.,

$$t \sim t', \quad \forall t, t' \in \mathcal{V}$$

2. Following the commutativity of distance functions, the utility function is also commutative, i.e.

$$u(t, t') = u(t', t), \quad \forall t, t' \in \mathcal{V}$$

Let the sensitivity of the utility function be $\Delta := \max_{t, t' \in \mathcal{V}} \max_{y \in \mathcal{V}} |u(t, y) - u(t', y)|$, the commutative property suggests that the difference in utility between any pair of outputs y and y' is also bounded, namely

$$|u(t_i, y) - u(t_i, y')| \leq \Delta, \quad \forall t_i, y, y' \in \mathcal{V}$$

Correspondingly, their output probabilities are also similar, as

$$\frac{\Pr[\mathcal{R}(t_i) = y]}{\Pr[\mathcal{R}(t_i) = y']} = \frac{\exp(\frac{\varepsilon}{2} \cdot \frac{u(t_i, y)}{\Delta})}{\exp(\frac{\varepsilon}{2} \cdot \frac{u(t_i, y')}{\Delta})} \leq e^{\varepsilon/2}.$$

With the additional constraint that $\sum_{y \in \mathcal{V}} \Pr[\mathcal{R}(t_i) = y] = 1$, we can derive:

$$\frac{e^{-\varepsilon/2}}{e^{-\varepsilon/2} + |\mathcal{V}| - 1} \leq \Pr[\mathcal{R}(t_i) = y] \leq \frac{e^{\varepsilon/2}}{e^{\varepsilon/2} + |\mathcal{V}| - 1},$$

$$\frac{e^{-\varepsilon/2}}{|\mathcal{V}|} < \Pr[\mathcal{R}(t_i) = y] < \frac{e^{\varepsilon/2}}{|\mathcal{V}|}.$$

Lemma 3.1. *Let $\mathcal{R}$ be an exponential mechanism for prompt tuning, then for any input t_i and any output y ,*

$$|\mathcal{V}| \cdot \Pr[\mathcal{R}(t_i) = y] \in (e^{-\varepsilon/2}, e^{\varepsilon/2})$$

²<https://github.com/sai4july/CusText>

³<https://github.com/mengtong0110/InferDPT>

⁴Yuan: If the high-utility outputs matter, it seems more natural to adopt a heavy-hitter histogram (i.e. putting all low-utility elements below certain threshold into one group). Also, if the analysis is not related to grouping strategy, maybe we can use min instead of mean for group utilities to disfavor low-utility outputs.

3.2 Sample Amplification

Let the input prompt be $(t_1, t_2, \dots, t_n)$. Suppose the warehouse contains a perturbed prompt $\mathcal{R}(t_1, \dots, t_n) = (y_1, y_2, \dots, y_n)$ using privacy budget ε , the target is to output $(y'_1, \dots, y'_n)$ which satisfies a more strict privacy requirement $\varepsilon' < \varepsilon$. Consider the following sample mechanism $\mathcal{S}$:

$$y'(t_i) = \begin{cases} \mathcal{R}(t_i), & \text{with probability } p, \\ \text{Random token from } \mathcal{V}, & \text{with probability } 1 - p. \end{cases}$$

For each (sensitive) token t_i , we output Y_i with probability p ; otherwise a random token is uniformly sampled from $\mathcal{V}$. We show that with a carefully decided p , the mechanism can achieve ε' -DP.

Lemma 3.2. *Let $\mathcal{R}$ be an EM-based mechanism with privacy budget ε . For*

$$p = \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon/2} - 1)(e^{\varepsilon' - \varepsilon/2} + 1)},$$

the mechanism $\mathcal{S} \circ \mathcal{R}$ satisfies ε' -DP.

Proof. Following the mechanism design, the probability of outputting any $y \in \mathcal{V}$ is

$$\Pr[\mathcal{S} \circ \mathcal{R}(t_i) = y] = p \cdot \Pr[\mathcal{R}(t_i) = y] + \frac{1 - p}{|\mathcal{V}|} = \frac{p \cdot |\mathcal{V}| \cdot \Pr[\mathcal{R}(t_i) = y] + (1 - p)}{|\mathcal{V}|}.$$

The privacy guarantee of $\mathcal{R}$ suggests that for any neighboring inputs $t_i \sim t'_i$ and any output $y \in \mathcal{V}$,

$$\Pr[\mathcal{R}(t_i) = y] \leq e^\varepsilon \cdot \Pr[\mathcal{R}(t'_i) = y].$$

Therefore, by Lemma 3.1 we have

$$\frac{\Pr[\mathcal{S} \circ \mathcal{R}(t_i) = y]}{\Pr[\mathcal{S} \circ \mathcal{R}(t'_i) = y]} = \frac{p \cdot |\mathcal{V}| \cdot \Pr[\mathcal{R}(t_i) = y] + (1 - p)}{p \cdot |\mathcal{V}| \cdot \Pr[\mathcal{R}(t'_i) = y] + (1 - p)} < \frac{1 + p(e^{\varepsilon/2} - 1)}{1 + p(e^{-\varepsilon/2} - 1)}, \quad (1)$$

Solve for $(1) \leq e^{\varepsilon'}$, we obtain

$$p \leq \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon/2} - 1) + (1 - e^{-\varepsilon/2})e^{\varepsilon'}} = \frac{e^{\varepsilon'} - 1}{(e^{\varepsilon/2} - 1)(e^{\varepsilon' - \varepsilon/2} + 1)}.$$

□

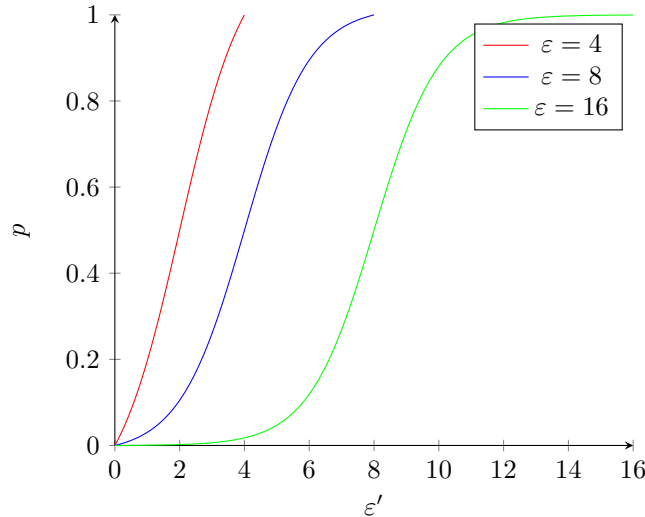


Figure 1: Mixing 0 and ε for ε'

Remarks. Figure 1 plots the probabilities for sampling different ε' 's. In particular, we have when $\varepsilon' = 0$, the mechanism degenerates into a uniform random sample; when $\varepsilon' = \varepsilon/2$, the sampling probability is exactly $p = 1/2$.

3.3 Generalization

We now consider a more general version of Lemma 3.3. Consider two perturbed prompt: $\mathcal{R}_1(t_i) = a_i$ with privacy budget ε_1 ; and $\mathcal{R}_2(t_i) = b_i$ with privacy budget ε_2 . Our target is to achieve ε' -DP where $\varepsilon_1 \leq \varepsilon' \leq \varepsilon_2$.

Similarly, consider the following sample mechanism $\mathcal{S}_2$: for each (sensitive) token t_i , output a_i with probability $1 - p$; or b_i otherwise. Denote the new mechanism by $\mathcal{M} := \mathcal{S}_2 \circ (\mathcal{R}_1, \mathcal{R}_2)$. Note that when $\varepsilon_1 = 0$, $\mathcal{M}$ degenerates into Lemma 3.3.

Lemma 3.3. *Let $\mathcal{R}_1, \mathcal{R}_2$ be two EM-based mechanisms with respective privacy budgets $\varepsilon_1 < \varepsilon_2$. For*

$$p = \frac{e^{\varepsilon'} - e^{\varepsilon_1}}{(e^{(\varepsilon_1 + \varepsilon_2)/2} - e^{\varepsilon_1}) + (1 - e^{(\varepsilon_1 - \varepsilon_2)/2})e^{\varepsilon'}},$$

the mechanism $\mathcal{M}$ satisfies ε' -DP.

Proof. Following previous analyses,

$$\Pr[\mathcal{M}(t_i) = y] = (1 - p) \cdot \Pr[\mathcal{R}_1(t_i) = y] + p \cdot \Pr[\mathcal{R}_2(t_i) = y].$$

Similarly, we solve for

$$\begin{aligned} \frac{\Pr[\mathcal{M}(t_i) = y_i]}{\Pr[\mathcal{M}(t'_i) = y_i]} &= \frac{(1 - p) \cdot \Pr[\mathcal{R}_1(t_i) = y] + p \cdot \Pr[\mathcal{R}_2(t_i) = y]}{(1 - p) \cdot \Pr[\mathcal{R}_1(t'_i) = y] + p \cdot \Pr[\mathcal{R}_2(t'_i) = y]} \\ &\leq \frac{e^{\varepsilon_1/2} + p(e^{\varepsilon_2/2} - e^{\varepsilon_1/2})}{e^{-\varepsilon_1/2} + p(e^{-\varepsilon_2/2} - e^{-\varepsilon_1/2})} \\ &= \frac{e^{\varepsilon_1} + p(e^{(\varepsilon_1 + \varepsilon_2)/2} - e^{\varepsilon_1})}{1 + p(e^{(\varepsilon_1 - \varepsilon_2)/2} - 1)} \leq e^{\varepsilon'} \end{aligned}$$

to get

$$p \leq \frac{e^{\varepsilon'} - e^{\varepsilon_1}}{(e^{(\varepsilon_1 + \varepsilon_2)/2} - e^{\varepsilon_1}) + (1 - e^{(\varepsilon_1 - \varepsilon_2)/2})e^{\varepsilon'}}$$

□

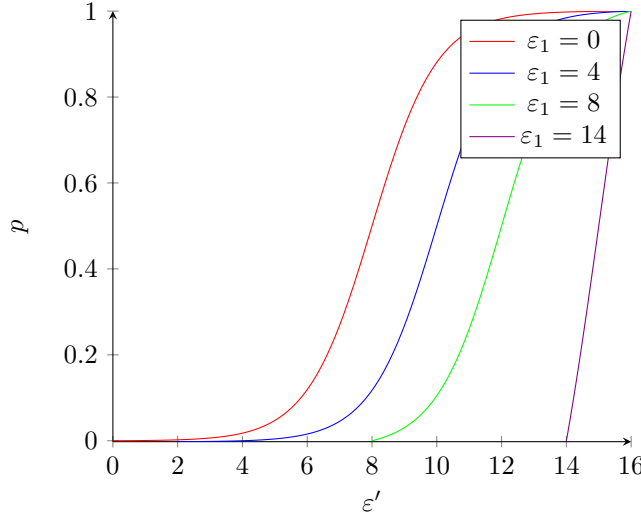


Figure 2: Mixing ε_1 and $\varepsilon_2 = 16$ for ε'

Remarks. Figure 2 the effect of the lower ε_1 . Note that when $\varepsilon_1 = 0$, $p = \frac{e^{\varepsilon'} - 1}{e^{\varepsilon_2/2} - 1 + e^{\varepsilon'} - e^{\varepsilon_2/2}}$ degenerates to Lemma 3.3. Some interesting results of ε' and p :

1. $\varepsilon' = \varepsilon_1$, then $p = 0$;
2. $\varepsilon' = \varepsilon_2$, then $p = 1$;
3. $\varepsilon' = (\varepsilon_1 + \varepsilon_2)/2$, then $p = 1/2$.

3.3.1 Detailed Analysis

Let $\mathcal{M} := \mathcal{S}_2 \circ (\mathcal{R}_1, \mathcal{R}_2)$ be the new mechanism. According to previous analysis, for any neighbouring instances $t_i \sim t'_i$ and any output y_i , we should have

$$\frac{\Pr[\mathcal{M}(t_i) = y_i]}{\Pr[\mathcal{M}(t'_i) = y_i]} = \frac{(1-p) \cdot \Pr[\mathcal{R}_1(t_i) = y_i] + p \cdot \Pr[\mathcal{R}_2(t_i) = y_i]}{(1-p) \cdot \Pr[\mathcal{R}_1(t'_i) = y_i] + p \cdot \Pr[\mathcal{R}_2(t'_i) = y_i]} \leq e^{\varepsilon'}. \quad (2)$$

Thus it suffices to set p such that

$$\max_{t_i, t'_i} \max_{y_i \in \mathcal{V}} \frac{(1-p) \cdot \exp(-\varepsilon_1/2 \cdot d(t_i, y_i)) \cdot C_{1,t_i} + p \cdot \exp(-\varepsilon_2/2 \cdot d(t_i, y_i)) \cdot C_{2,t_i}}{(1-p) \cdot \exp(-\varepsilon_1/2 \cdot d(t'_i, y_i)) \cdot C_{1,t'_i} + p \cdot \exp(-\varepsilon_2/2 \cdot d(t'_i, y_i)) \cdot C_{2,t'_i}} \leq e^{\varepsilon'}.$$

3.4 Generalization to GP

GP takes $\Pr[\mathcal{R}(x) = y] \propto e^{-\varepsilon d(x,r)/2}$ and satisfies

$$\frac{\Pr[\mathcal{R}(x) = y]}{\Pr[\mathcal{R}(x') = y]} \leq e^{\varepsilon \cdot d(x, x')}$$

Let $d = d(x, x')$ then the SA wants to get

$$\frac{\Pr[\mathcal{M}(x) = y]}{\Pr[\mathcal{M}(x') = y]} = \frac{p \cdot e^{\varepsilon_1 d} q_1 + (1-p) \cdot e^{\varepsilon_2 d} q_2}{p \cdot q_1 + (1-p) \cdot q_2} \leq e^{\varepsilon' d}$$

We can derive the same result as,

$$p \leq \frac{(e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}{(e^{\varepsilon_1} - e^{\varepsilon'}) \cdot q_1 + (e^{\varepsilon'} - e^{\varepsilon_2}) \cdot q_2}$$

But we do not have the same inequality for q_1 and q_2 . Only $e^{-\varepsilon_1 d_{\max}/2} < q_1 < e^{-\varepsilon_2 d_{\max}/2}$. And $q_1/q_2 > e^{-(\varepsilon_1 + \varepsilon_2) d_{\max}/2}$.

Can we calculate $q_{1\max}$ and $q_{2\min}$? Compute it through the matrix?

4 Our System

Algorithm 1: DP Perturbed Prompts Warehouse

Protocol: Offline Prepare

Input: Plaintext prompt $x = (t_1, t_2, \dots, t_n)$, privacy parameters $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_k)$, vocabulary $\mathcal{V}$

Output: Perturbed prompts $(\mathcal{X}^0, \mathcal{X}^1, \dots, \mathcal{X}^{k+1})$

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1  $\mathcal{X}^0 \leftarrow (\text{Rand}(\mathcal{V}))^n$ ; //  $\varepsilon_0 = 0$ 
2 for  $i \leftarrow 1$  to  $k$  do
3    $\mathcal{X}^i \leftarrow \text{BaseDPPerturb}(x, \varepsilon_i)$ ;
4  $\mathcal{X}^{k+1} \leftarrow x$ ; //  $\varepsilon_{k+1} = \infty$ 
5 return  $\mathcal{X}$ 
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Protocol: Online Query

Input: Privacy parameter ε

Output: Perturbed prompt

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6  $i \leftarrow$  the ID such that  $\varepsilon_i \leq \varepsilon < \varepsilon_{i+1}$ ;
7  $(a_1, a_2, \dots, a_n), (b_1, b_2, \dots, b_n) \leftarrow \mathcal{X}^i, \mathcal{X}^{i+1}$ ;
8 if  $i = k$  then
9    $p \leftarrow 1 - \frac{1}{(e^\varepsilon - e^{\varepsilon_i}) \cdot e^{\varepsilon_i/2}}$ ;
10 else
11    $p \leftarrow \frac{e^\varepsilon - e^{\varepsilon_i}}{(e^{\varepsilon_{i+1}} - e^\varepsilon) \cdot e^{-(\varepsilon_i + \varepsilon_{i+1})/2} + (e^\varepsilon - e^{\varepsilon_i})}$ ;
12 for  $j \leftarrow 1$  to  $n$  do
13    $y \leftarrow \text{Ber}(p)$ ;
14    $t_j \leftarrow a_j$  if  $y = 1$  else  $b_j$ ;
15 return  $(t_1, t_2, \dots, t_n)$ 
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In our DP perturbed prompts warehouse, we perturb each prompt x with some pre-defined privacy budgets $\varepsilon_1 < \varepsilon_2 < \dots < \varepsilon_k$, and store them $\mathcal{X}^i = \mathcal{R}(x, \varepsilon_i)$ in the warehouse. We all add two special case: $\varepsilon_0 = 0$ which stores $\mathcal{X}^0$ as a random choice of $\mathcal{V}$ of length $|x|$, and $\varepsilon = \infty$ which stores the original

Once a user asks for prompts with privacy budget ε (In traditional DP warehouse [?], this is usually related to the money she pays for.), we search the warehouse to find $\varepsilon_l, \varepsilon_r$ such that $\varepsilon_l \leq \varepsilon \leq \varepsilon_r$. Then we calculate p and use our sampling mechanism to output a new perturbed prompt satisfying ε -DP.

In this case, we only need to perform n Bernoulli samples. It is much more efficient than re-compute a new perturbed prompt with ε , but can (may) reach the same accuracy as re-compute ones; (must) have better accuracy than the fixed budget method.

Quick response time and light computation workload, no embedding model needed.

High accuracy level under the same privacy guarantee.

4.1 Secure Version

It can be implemented in “secure outsourced computation” framework. Generally speaking, we encrypt the warehouse and compute the sampling mechanism under MPC/TEE. Since ε is public information, the whole computation procedure is also MPC/TEE-friendly.

References

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